AuraHealth: first-aid guides for clinical review

AuraHealth is a free research prototype (simulation only, not a medical device). While someone waits for an ambulance, it shows these steps one at a time and reads them aloud, and it saves the time of each step for the doctor. The steps are fixed text written in plain English from the **International Federation of Red Cross and Red Crescent Societies (IFRC)**, *International first aid, resuscitation and education guidelines 2020*. No AI writes or changes them. For teenagers and adults. Babies and young children need different steps: tell the 112 operator the age. Medicine doses are left out on purpose (for example aspirin for chest pain). Ask the 112 operator. The emergency number shown is 112 (India).

Please check each step: is it correct, safe and clear for an ordinary person under stress? Tick OK, Change or Remove, and write the change in Comments. Anything missing that a bystander should do (or must not do), please add.

1. Someone collapsed: start here

When: You are not sure what is wrong. · **IFRC 2020 pages:** 128-139

#	Type	What the app says (read aloud)	Your check	Comments
1	Step	Make sure the place is safe for you, for example from traffic, fire or electricity. Go to the person only when it is safe.	☐ OK ☐ Change ☐ Remove	
2	Question	Shout their name and tap their shoulders. Do they respond? Answers: Yes, they respond / No response	☐ OK ☐ Change ☐ Remove	
3	Question	Look at their chest and listen. Are they breathing normally? Gasping, noisy or very slow breathing is NOT normal. Answers: Yes, normally / No, or only gasping / I'm not sure	☐ OK ☐ Change ☐ Remove	

Good to know

- If you are not sure whether they are breathing normally, start CPR. That is what the Red Cross guidelines advise.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

2. Not responding and not breathing normally

When: No response to voice or touch, and no breathing or only gasping. · **IFRC 2020 pages:** 134-146, 397

#	Type	What the app says (read aloud)	Your check	Comments
1	Call	Call 112 now, or ask someone nearby to call. Put the phone on speaker so your hands are free.	☐ OK ☐ Change ☐ Remove	
2	Step	Kneel beside them. Put the heel of one hand in the middle of the chest, the other hand on top, fingers locked together.	☐ OK ☐ Change ☐ Remove	
3	CPR beat	With straight arms, push hard, straight down, about 5 cm, then let the chest come all the way back up. Push on every beat: 100 to 120 times a minute.	☐ OK ☐ Change ☐ Remove	
4	Step	Keep going without stopping until help takes over, or they cough, open their eyes, move on purpose or breathe normally.	☐ OK ☐ Change ☐ Remove	

Do not

- Do not stop to look for a pulse. A pulse check is unreliable and wastes time.
- Do not stop compressions to wait for the ambulance.

Good to know

- If a defibrillator (AED) is nearby, ask someone to bring it, switch it on and follow its voice.
- If someone else can take over, swap so you don't get too tired, with as little pause as possible.
- If you are carrying the person to hospital, keep CPR going on a firm surface.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

3. Not responding but breathing normally

When: No response to voice or touch, but normal, regular breathing. · **IFRC 2020 pages:** 128-133

#	Type	What the app says (read aloud)	Your check	Comments
1	Step	Roll them onto their side and gently tilt their head back. This keeps the airway open and stops them choking if they vomit.	☐ OK ☐ Change ☐ Remove	
2	Call	Call 112. Someone who does not respond needs help, even if they breathe.	☐ OK ☐ Change ☐ Remove	
3	Step	Stay with them and keep checking their breathing. If they stop breathing normally, start CPR.	☐ OK ☐ Change ☐ Remove	

Do not

- If you think they hurt their neck or back, do not roll them. Keep the airway open by tilting the head and lifting the chin.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

4. Choking

When: Something is stuck in the throat. · **IFRC 2020 pages:** 170-174, 400

#	Type	What the app says (read aloud)	Your check	Comments
1	Question	Can they cough, speak or breathe? Answers: Yes / No	☐ OK ☐ Change ☐ Remove	
2	Step	Tell them to keep coughing. Stay with them and watch closely: it can get worse.	☐ OK ☐ Change ☐ Remove	
3	Step	Stand behind them, bend them forward, and give up to 5 sharp blows between the shoulder blades with the heel of your hand.	☐ OK ☐ Change ☐ Remove	
4	Step	Still blocked? Stand behind, put a fist between the belly button and the ribs, hold it with your other hand, and pull sharply in and up, up to 5 times.	☐ OK ☐ Change ☐ Remove	
5	Call	Still blocked? Call 112 now. Keep repeating 5 back blows and 5 abdominal thrusts.	☐ OK ☐ Change ☐ Remove	

Do not

- Do not sweep inside the mouth blindly with your fingers: it can push the object further down. Only take out something you can see.

Good to know

- Babies under one year need different steps. Tell the 112 operator the age.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

5. Heavy bleeding

When: Blood is flowing fast from a wound. · **IFRC 2020 pages:** 187-194

#	Type	What the app says (read aloud)	Your check	Comments
1	Step	Press hard on the wound. Use a cloth, bandage or plastic between your hand and the blood. If they can, they can press it themselves.	☐ OK ☐ Change ☐ Remove	
2	Step	Help them lie down, so they do not faint and fall.	☐ OK ☐ Change ☐ Remove	
3	Call	Call 112, or ask someone to call while you keep pressing.	☐ OK ☐ Change ☐ Remove	
4	Step	If blood soaks through, put another cloth on top and press harder. Do not lift the first cloth.	☐ OK ☐ Change ☐ Remove	
5	Step	Only if pressing does not stop heavy bleeding from an arm or leg, and you have a tourniquet: put it on above the wound. The time is saved for the doctor.	☐ OK ☐ Change ☐ Remove	
6	Step	Keep them lying down and warm, for example covered with clothes or a blanket.	☐ OK ☐ Change ☐ Remove	

Do not

- If an object such as a knife is stuck in the wound, do not pull it out. Press around it.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

6. Chest pain

When: Pain or pressure in the chest that does not go away. · **IFRC 2020 pages:** 264-270

#	Type	What the app says (read aloud)	Your check	Comments
1	Step	Help them sit down in a comfortable position. If a doctor prescribed them heart medicine for this, help them take it.	☐ OK ☐ Change ☐ Remove	
2	Call	Call 112 now. Ask someone to bring a defibrillator (AED) if there is one nearby.	☐ OK ☐ Change ☐ Remove	
3	Step	Stay with them, keep them calm, and watch their breathing and whether they respond. If they stop responding and are not breathing normally, start CPR.	☐ OK ☐ Change ☐ Remove	

Good to know

- Women, older people and people with diabetes may have milder or unusual signs. If in doubt, treat it as serious.
- Medicine doses are not given here. Ask the 112 operator what to give.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

7. Signs of a stroke

When: Sudden weak face or arm, or strange speech. · **IFRC 2020 pages:** 271-278

#	Type	What the app says (read aloud)	Your check	Comments
1	Question	Ask them to smile. Does one side of the face droop? Answers: Yes / No / Not sure	☐ OK ☐ Change ☐ Remove	
2	Question	Ask them to raise both arms. Does one arm drift down or not move? Answers: Yes / No / Not sure	☐ OK ☐ Change ☐ Remove	
3	Question	Ask them to repeat a simple sentence. Is their speech slurred or strange? Answers: Yes / No / Not sure	☐ OK ☐ Change ☐ Remove	
4	Call	Any of these signs? Call 112 now. Tell them when the signs started, or the last time the person was seen well. If you don't know, say so.	☐ OK ☐ Change ☐ Remove	
5	Step	Help them into a comfortable position and support them so they don't fall. Keep talking to them and watch their breathing.	☐ OK ☐ Change ☐ Remove	

Do not

- Do not give them anything to eat or drink: swallowing may not be safe.

Good to know

- Even if the signs go away, they still need a doctor quickly.
- Very high or low blood sugar can look similar. Tell the doctor about diabetes.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

8. Snakebite

When: Bitten by a snake, or you think so. · **IFRC 2020 pages:** 254-257

#	Type	What the app says (read aloud)	Your check	Comments
1	Step	Help them lie down and keep as still as possible. Moving makes the venom spread faster.	☐ OK ☐ Change ☐ Remove	
2	Call	Call 112. They need a hospital that has antivenom.	☐ OK ☐ Change ☐ Remove	
3	Step	Take off rings, bangles, watches and tight clothes near the bite before it swells. Move the limb as little as possible.	☐ OK ☐ Change ☐ Remove	
4	Step	If help is far away, keep the bitten arm or leg still with a cloth or non-stretchy bandage wrapped around it. Do not wrap it tight.	☐ OK ☐ Change ☐ Remove	
5	Step	Do not let them walk on a bitten leg. Carry them, for example two people making a chair with their arms.	☐ OK ☐ Change ☐ Remove	
6	Step	Keep watching their breathing and whether they respond. If it is safe, take a photo of the snake from a distance.	☐ OK ☐ Change ☐ Remove	

Do not

- Do not tie a tight band (tourniquet) around the limb.
- Do not cut the bite or suck out the venom.
- Do not put ice on it or rub it.
- Do not try to catch or kill the snake.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

9. Seizure (fit)

When: Shaking, stiffening or not responding with twitching. · **IFRC 2020 pages:** 295-298

#	Type	What the app says (read aloud)	Your check	Comments
1	Step	Tap Done now: the seizure start time is saved for the doctor.	☐ OK ☐ Change ☐ Remove	
2	Step	Move hard or sharp things away. Do not move them unless they are in danger, for example on a road.	☐ OK ☐ Change ☐ Remove	
3	Step	Put something soft under their head. Take off glasses and loosen anything tight around the neck.	☐ OK ☐ Change ☐ Remove	
4	Call	Call 112 if: it is their first seizure, it lasts more than 5 minutes, they are hurt, they don't wake up after, it happens again, they are pregnant or diabetic, or you are not sure.	☐ OK ☐ Change ☐ Remove	
5	Step	When the shaking stops, tap Done: the stop time is saved.	☐ OK ☐ Change ☐ Remove	
6	Question	Now check their breathing. Are they breathing normally? Answers: Yes, normally / No, or only gasping	☐ OK ☐ Change ☐ Remove	

Do not

- Do not hold them down.
- Do not put anything in their mouth or between their teeth.
- Do not give them food, drink or medicine until they are fully awake.

Overall for this guide: ☐ Approve ☐ Approve with the changes above ☐ Do not use

Reviewer

Name	
Qualification / registration	
Date	
May we say on our demo video that you reviewed these steps? (Yes / No)	
Signature	